Anindya Sen

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Education

Vellore Institute of Technology (VIT) *Vellore, Tamil Nadu, India*
B.Tech. in Computer Science and Engineering
CGPA: 8.01/10.0

Relevant Courses: Applied Linear Algebra, Artificial Intelligence, Web Mining, Image Processing, Machine Learning, Applications of Differential Equations, Parallel and Distributed Computing, Internet of Things, Human Computer Interaction, Digital Forensics, Information Security Management.

Research Interests

Natural Language Processing	Machine Learning	Image Processing
Data Mining	Computer Forensics	Computer Vision

Publication & Research Experience

- **Journal Publications**
  - Sarkar, S.S., **Sen, A.**, Krishnamoorthy, A. *et al.* **Route Planning Service for Emergency Vehicles with Increased Accuracy and Efficiency for Online Platforms**. *SN COMPUT. SCI.* **3**, 400 (2022). https://doi.org/10.1007/s42979-022-01289-5
- **Book Chapters**
  - Dey, K., Chowdhury, F. K., **Sen, A.**, Arka, A. M., Chowdhury, O., Roy, H., & Rahman, M. Z. (2024). **Graphene and its hybrid materials: Properties and applications.** Reference Module in Materials Science and Materials Engineering. https://doi.org/10.1016/b978-0-323-96020-5.00214-4
- **Undergraduate Final Semester Capstone Project**
  - **Title:** Decentralized Healthcare Network Platform
  - **Supervisor:** Dr. Krishnamoorthy A.
  - **Description:** Using state-of-the-art blockchain technology, we have built a totally decentralized system in which data is saved in our proprietary file system called IHFS, which is entirely protected from data security and integrity concerns. (Link)

Standardized Test Score

IELTS: 8.0 (Overall)	Listening: 8.5	Reading: 9.0	Writing: 7.0	Speaking: 7.5
GRE: 312	Quant: 158	Verbal: 154	AWA: 3.0	

Work Experience

IT Engineer

Pacific Casuals Ltd., Chattogram, Bangladesh
Feb 2025 – Dec 2025

- Providing basic IT support (hardware, software, network)
- Installing and maintaining computer systems
- Troubleshooting technical issues
- Assisting with network setup and connectivity
- Helping staff with IT-related problems

Skills

- **Programming Languages:** Python, C, C++, Java, JavaScript.
- **Frameworks and Libraries:** Tensorflow, PyTorch, Keras, OpenCV, Numpy, Pandas, Scikit-learn, Django, Node.JS.

Projects

- "Object" a Web Based Object Detection Site (Link)
    - Developed a real-time object detection web application using TensorFlow.js and the pre-trained COCO-SSD model. Implemented client-side inference for detecting multiple objects in images/video directly in the browser. Designed an interactive UI using JavaScript, HTML, and CSS for live detection with low latency and efficient performance.

- Offensive Language Identification System (Link)
    - Developed a web-based NLP application to detect offensive language using the OffensEval (SemEval 2019 Task 6) dataset. Built the system with Flask (backend) and HTML, CSS, and JavaScript (frontend), allowing users to submit text or upload files for analysis. Implemented and compared Naive Bayes and LSTM models, achieving 71% and 75% accuracy respectively. Evaluated performance trade-offs between speed (0.01s for Naive Bayes) and accuracy (LSTM), improving overall model selection and system efficiency.

- Comparing word search technique between Sequential and Parallel Method (Link)
    - Developed and evaluated word search algorithms for large text datasets using C/C++ and OpenMP, implementing both sequential and multithreaded approaches. Designed a parallel solution to process multiple files concurrently, significantly reducing execution time (1.34s to 0.10s). Analyzed performance, scalability, and efficiency, demonstrating clear speedup and improved resource utilization in the parallel model.

- Real-time 3D Product Display and Demo using ARCore
    - Developed an Android application using Java and ARCore to enable real-time 3D product visualization in augmented reality. Implemented markerless AR to place and interact with virtual objects in a real-world environment. Designed an interactive user interface for seamless product demonstration, enhancing user engagement and visualization experience.

- Efficient Image Encryption Algorithm
    - Developed a Python-based image encryption system using matrix manipulation techniques to transform pixel values and secure image data. Implemented reversible encryption–decryption processes ensuring data integrity while maintaining low computational overhead.

- Detection of Motorcyclists and Construction workers without Helmet from Live Surveillance Video
    - Developed a real-time surveillance system using YOLO-based CNN models to detect individuals without helmets in live video streams. Implemented object detection to identify motorcyclists and construction workers, followed by classification to determine helmet usage.

- Detecting and Tracking Irregular Objects Based on Their Shape and Color In Real-Time
    - Developed a computer vision system to detect and track irregularly shaped objects in real time using OpenCV and NumPy. Implemented color-based segmentation and contour detection to identify objects, followed by tracking across video frames.

- Garage Management System
    - Developed a garage management system in C++ to simulate resource allocation and prevent deadlock scenarios in operating systems. Implemented synchronization and deadlock avoidance techniques to manage vehicle servicing efficiently. Demonstrated concepts of resource sharing and process coordination.

- Pharmacy Management System
    - Developed a database-driven web application using PHP, MySQL, and HTML to manage pharmacy operations. Implemented features for inventory management, billing and record keeping, enabling efficient tracking of medicines and transactions. Designed a structured database to ensure data consistency and reliable system performance.

- Servo Motor Speed Monitor (Link)
    - Developed an IoT-based servo motor speed monitoring system using Proteus for circuit simulation and Blynk for real-time data visualization. Implemented virtual sensors and microcontroller logic to measure motor speed and transmit data to a cloud interface for remote monitoring and control.

- Telemedicine and Monitoring App for Elderly People
    - Conceptualized a telemedicine and remote monitoring solution as part of a Lean Startup project, focusing on elderly care. Designed system architecture, conducted cost analysis and developed marketing and go-to-market strategies. Created a detailed blueprint and prototype design to demonstrate feasibility and user workflow.

Activities

- **IEEE Power and Energy Society (IEEE PES), VIT Chapter**
  Core Committee Member *December 2018-December 2020*
    - Conducted Workshops on: Machine Learning, MATLAB.
    - Represented in: Gravitas 2019 and Riviera 2019.

Achievements & Awards

- Government Scholarship for Secondary School Certificate Examination
- Government Scholarship for Junior School Certificate Examination (2nd position in Chattogram Board)
- Achieved awards from various art competitions.

MOOCs & Certifications

- Advanced Algorithms and Complexity (Link)
- The Arduino Platform and C Programming (Link)
- Fundamentals of Parallelism on Intel Architecture (Link)
- Essential of Machine Learning (Link)
- Logistics and Supply Chain Management (Link)
- Marketing Management (Link)
- Web Application Technologies and Django (Link)